

INTRODUCTION TO APPLIED CRYPTOGRAPHY

Cryptography, old and new

- ◆ Crypto is an ancient discipline
- ◆ Recall:
 - Julius Caesar
 - Enigma machine
- ◆ Cryptography as a science (modern cryptography) has a short, but exciting history. Most of it has occurred within the last **40 years**

This course is an introduction to
modern cryptography



Main Goals of Cryptography

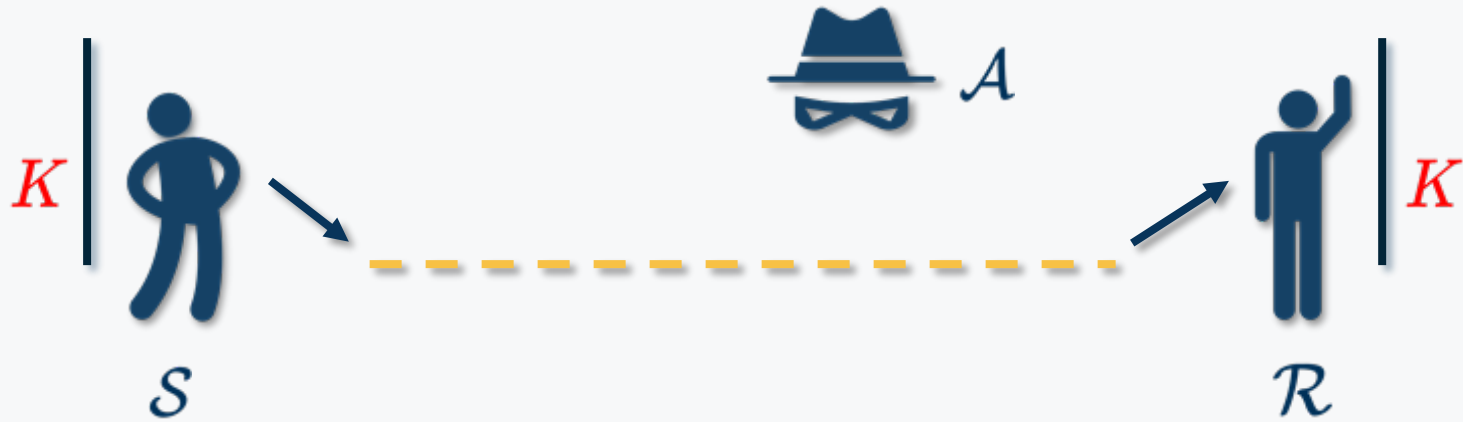
- ◆ Data Privacy
- ◆ Data Authenticity
 - message came from where it claims
- ◆ Data Integrity
 - message has not been modified on the way

When is Cryptography used?

- Online shopping and banking
- Using a cell phone
- Paying with credit cards
- ...

Half of the internet is now encrypted!

Players and Settings



1. Symmetric-key setting

Players and Settings



...	...
$\mathcal{R}$	$pk_{\mathcal{R}}$
...	...

2. Asymmetric (public)-key setting

Goals and Primitives (tools)

goal \ setting	symmetric-key	asymmetric-key
data privacy	symmetric (secret-key) encryption	asymmetric (public-key) encryption
data authenticity/integrity	message authentication code (MAC)	digital signature scheme

Course Objectives

- We will study the practical solutions for the basic cryptographic goals in each setting, e.g.
 - AES-based modes of operation
 - HMAC
 - RSA-OAEP encryption
 - RSA and DSA signatures
 - ...
- We will also learn the more fundamental principles of:
 - What is a good scheme
 - How to estimate and compare security

How good is a Scheme?

◆ “Trial and error” approach

1. Try to find an attack
2. If an attack is found, the scheme is insecure.
Fix the scheme and repeat step 1
3. If no attack is found then...?

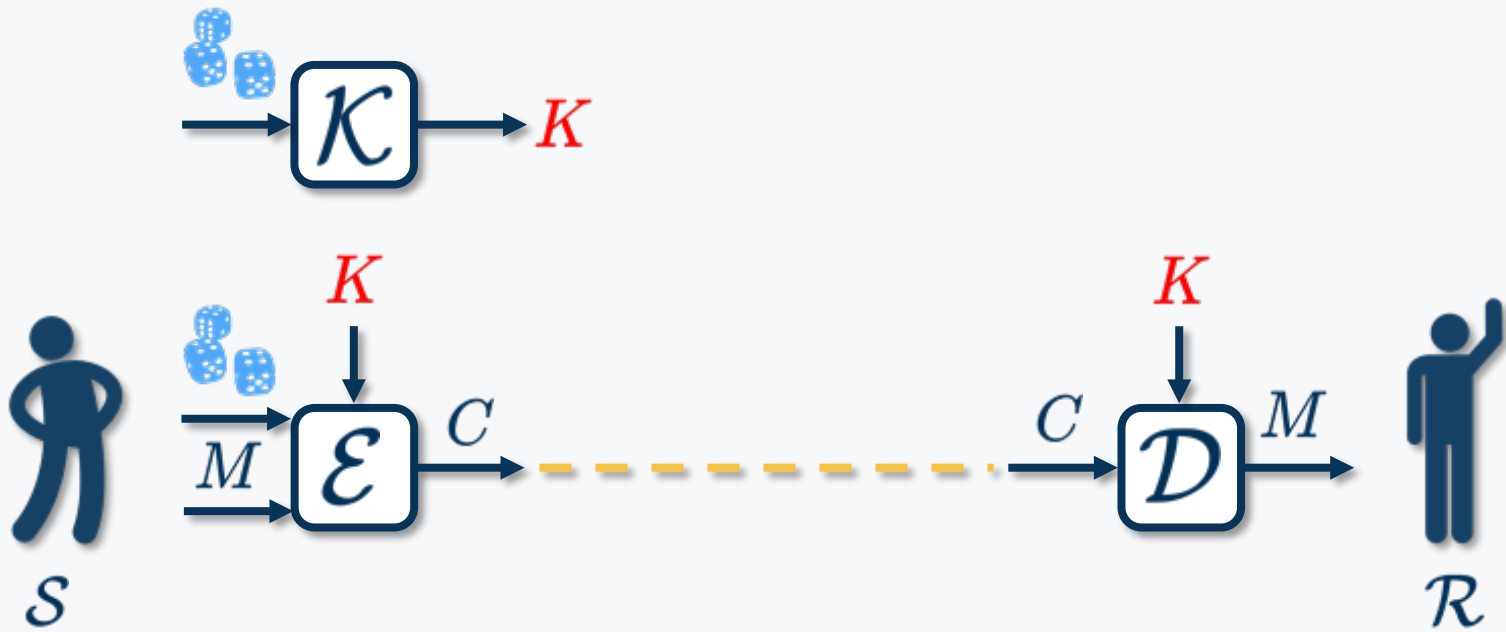
◆ “Provable security” approach

- Show that if an attack is found (a scheme is insecure), then one can break some trusted assumptions (e.g. factoring)
- Requires a definition of what “secure” means

SYMMETRIC ENCRYPTION

Syntax of Symmetric Encryption

- ◆ A symmetric encryption scheme consists of descriptions of ...
 - The message space $\mathcal{MsgSp}$
 - The key generation algorithm $\mathcal{K}$ (or the key space $\mathcal{KeySp}$)
 - The encryption algorithm $\mathcal{E}$
 - The decryption algorithm $\mathcal{D}$



It is required that for every $M \in \mathcal{MsgSp}$ and every $K \in \mathcal{KeySp}$ (or output by $\mathcal{K}$), $\mathcal{D}(K, \mathcal{E}(K, M)) = M$

OneTimePad

◆ OneTimePad = $(\mathcal{K}, \mathcal{E}, \mathcal{D})$, $\text{MsgSp} = \{0, 1\}^n$

$\mathcal{K}$ return a random n -bit string K $\text{KeySp} = \{0, 1\}^n$

$\mathcal{E}(K, M) : C \leftarrow M \oplus K$ return C

$\mathcal{D}(K, C) : M \leftarrow C \oplus K$ return M

◆ Example

$M = 011111111011101$

$K = 110010011010100$

$C = 101101100001001$

Intuition for Security

- ◆ Ciphertexts do not give the adversary ANY new information

Perfect (Shannon) Security

- ◆ Def: an encryption scheme $\mathcal{SE} = (\mathcal{K}, \mathcal{E}, \mathcal{D})$ is Shannon-secure, if for every ciphertext C and messages M_1, M_2

$$\Pr[\mathcal{E}(K, M_1) = C] = \Pr[\mathcal{E}(K, M_2) = C]$$

Theorem

- OneTimePad is a perfect-secure encryption scheme

Proof

- Fix any ciphertext $C \in \{0, 1\}^n$
- For every M , $\Pr[\mathcal{E}(K, M) = C] = \Pr[K = M \oplus C] = 2^{-n}$

Theorem (*Shannon's theorem, optimality of OneTimePad*)

- ◆ If a scheme is Shannon-secure, then $|\mathcal{KeySp}|$ cannot be smaller than $|\mathcal{MsgSp}|$

Proof

- ◆ Fix a ciphertext C (by picking M_1, K and setting $C = \mathcal{E}(K, M_1)$)

Thus $\Pr[\mathcal{E}(K, M_1) = C] > 0$

- Assume there exists M_2 such that $\Pr[\mathcal{D}(K, C) = M_2] = 0$
- By the correctness requirement $\Pr[\mathcal{E}(K, M_2) = C] = 0$
- Therefore $\Pr[\mathcal{E}(K, M_1) = C] \neq \Pr[\mathcal{E}(K, M_2) = C]$ that violates Shannon secrecy
- Thus for every $M_2 \in \mathcal{MsgSp}$ there exists $K' \in \mathcal{KeySp}$
- $\mathcal{D}(K', C) = M_2$ and $|\mathcal{KeySp}| \geq |\mathcal{MsgSp}|$

- So we cannot do better than the OneTimePad, but we know it is impractical (why?). Is it the end?

YES, of the “information-theoretic” crypto

NO, if we relax the security requirement, and assume that adversaries are computationally bounded

- We will also assume that...

- There are some “hard” problems
- Secret keys are kept secret
- Algorithms are public (Kerckhoff’s principle)

We move to the area of “**computational-complexity**” crypto, that opens many possibilities